- **Internet Protocol (IP):** Routable protocol which uses IP addresses to deliver packets. It is an unreliable protocol, does not guarantee delivery of information. **Address Resolution Protocol (ARP):** Resolves IP addresses to MAC (Medium Access Control) addresses. (A MAC address is a hardware identification number that uniquely identifies each device on a network.)i.e., to map IP network addresses to the hardware addresses. **Internet Control Message Protocol (ICMP):** Used by network devices to send error messages and operational information. Example: A host or router might not be reached or a request service is not presented.
- **Internet Group Management Protocol (IGMP):** It is a communication protocol used by hosts and routers to send Multicast (group Communication) messages to multiple IP addresses at once.

Transport Layer

The sessions are recognized and data packets are swapped between hosts in this layer. Two main protocols established at this layer are:

- **Transmission Control Protocol (TCP):** Provides reliable connection oriented transmission between two hosts. It ensures delivery of packets between the hosts. **User Datagram Protocol (UDP):** Provides connectionless, unreliable, one-to-one or one-to-many delivery.

Application Layer

The Application layer of the TCP/IP model is similar to the Session, Presentation, and Application layers of the OSI Reference Model. The most popular Application layer protocols are:

Hypertext Transfer Protocol (HTTP): The core protocol of the World Wide Web. **File Transfer Protocol (FTP):** enables a client to send and receive complete files from a server. **Telnet:** connect to another computer on the Internet. **Simple Mail Transfer Protocol (SMTP):** Provide e-mail services. **Domain Name System (DNS):** Refer to other host computers by using names rather than numbers.

POINTS TO REMEMBER

- The Internet is a network of global connections – comprising private, public, business, academic and government networks – linked by guided, wireless and fiber-optic technologies.
- ARPANET was Advanced Research Projects Agency (ARPA) of the U.S. government in 1969 and was first recognized
- **INTRANET:** It is a private network within an enterprise to share company data and computing resources between the employees.
- **EXTRANET:** It is a private network that uses Internet technology and the public telecommunication system to securely share business's information with suppliers, vendors, partners, customers, or other businesses.

XII CompApp_English version CHAPTER 11.indd 158 23-02-2020 12:39:10